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LECTURE NOTES

UCHUT347 - ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

MODULE 4: RENEWABLE ENERGY AND SUSTAINABLE TECHNOLOGIES

I. RENEWABLE ENERGY AND SUSTAINABLE TECHNOLOGIES

Overview of Renewable Energy Sources

1. Solar Energy

- **Photovoltaic (PV) cells**
  - PV cells convert sunlight directly into electricity using semiconductor materials (commonly silicon).
  - Widely used in rooftop solar, solar streetlights, solar farms.
  - Example: **Rewa Solar Park (750 MW), Madhya Pradesh, India.**
- **Solar Thermal Systems**
  - Concentrated Solar Power (CSP) plants use mirrors to focus sunlight on a receiver, generating heat which is then used to produce electricity via turbines.
  - Example: **CSP plants in Rajasthan desert region.**

2. Wind Energy

- **Onshore Wind Turbines**
  - Installed on land, common in regions with steady winds.
  - Example: **Muppandal Wind Farm, Tamil Nadu (~1,500 MW)**.
- **Offshore Wind Turbines**
  - Placed in coastal waters where winds are stronger and steadier.
  - Example: **UK's Hornsea Project** is the largest offshore wind farm.

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3. Hydropower

- **Small-Scale Hydropower (≤25 MW)**
  - Less ecological impact, often "run-of-river" type without large reservoirs.
  - Example: Kerala's small hydro projects in Idukki and Wayanad.
- **Large-Scale Dams**
  - Examples: **Three Gorges Dam (China)**, **Bhakra Nangal (India)**.
  - Provide irrigation, flood control, and energy but cause displacement and ecological damage.

4. Biomass & Biogas

- **Biomass Energy**: Conversion of crop residues, forestry waste, and municipal solid waste into energy.
- **Biogas**: Produced through anaerobic digestion of organic waste, yielding methane-rich gas.
- Applications: Cooking fuel, small-scale power generation, transport fuel (bio-CNG).
- Example: **India's National Biogas and Manure Management Programme (NBMMP).**

5. Emerging Sources

- **Tidal Energy**: Harnesses energy from tidal waves; predictable but site-specific.
- **Geothermal Energy**: Uses heat from Earth's interior; common in volcanic regions.
- **Green Hydrogen**: Produced by splitting water using renewable electricity (electrolysis).
  - Seen as the **fuel of the future** for transport, steel, and chemical industries.
  - Example: India's **National Green Hydrogen Mission (2023)**.

Sustainable Technologies in Energy

1. Smart Grids & Energy Storage

- Smart grids integrate renewable energy with digital monitoring to balance demand and supply.
- **Energy storage solutions**:
  - Lithium-ion battery banks (for solar/wind farms).

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- Pumped hydro storage (excess energy used to pump water uphill, released to generate power later).
- Hydrogen storage for long-term backup.

2. Building-Integrated PV (BIPV)

- Solar panels integrated into walls, roofs, or windows of buildings.
- Saves land space, reduces building energy consumption.
- Example: Glass solar façades in smart city projects.

3. Distributed Generation

- Small-scale power generation close to consumers (e.g., rooftop solar, micro-wind turbines).
- Reduces transmission losses and increases resilience.
- Example: Kerala's **Soura rooftop solar program**.

4. Hybrid Systems

- Combine different renewable sources for reliability.
  - **Solar + Wind** → Complements day-night variability.
  - **Solar + Biomass** → Continuous power with biofuel backup.
- Example: **Hybrid solar-wind farms in Gujarat.**

Challenges and Opportunities

Challenges

1. **Intermittency**
   - Solar power unavailable at night; wind varies seasonally.
   - Requires large-scale storage solutions.
2. **High Initial Capital Cost**
   - Installation of solar farms, offshore wind, hydrogen electrolysis plants is expensive.
   - Long-term, however, costs are falling (solar is already cheaper than coal in many regions).

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3. **Land-Use Conflicts**
    - Solar parks occupy agricultural land; wind farms affect local ecosystems.
    - Example: Farmers in Tamil Nadu protesting against loss of grazing land for wind farms.

Opportunities

1. **India's Ambitious Target (450 GW renewable by 2030)**
    - Includes solar, wind, biomass, and small hydro.
    - Aims to reduce fossil fuel dependency and meet Paris Agreement goals.
2. **Kerala's Solar-Powered Water Metro**
    - Kochi Water Metro is world's first solar-powered urban water transport system.
    - Integrates clean transport with renewable energy.
3. **Global Shift to Green Hydrogen Economy**
    - Countries like Germany, Japan, and India investing in hydrogen production.
    - Hydrogen can decarbonize steel, cement, and shipping industries—sectors hard to electrify

II. CLIMATE CHANGE AND ENGINEERING SOLUTIONS

1. Basics of Climate Science

What is Climate?

- **Climate** refers to the **long-term average weather conditions** in a region, typically studied over 30 years or more (temperature, rainfall, wind patterns).
- Distinct from **weather**, which is short-term atmospheric conditions.

Greenhouse Effect & Greenhouse Gases (GHGs)

- Earth receives energy from the sun in the form of short-wave radiation.
- Some energy is reflected, while most is absorbed by land and oceans.
- Earth re-radiates heat as long-wave infrared radiation.
- **Greenhouse gases (GHGs)** like **CO_2, CH_4 (methane), N_2O (nitrous oxide), and water vapor** trap this heat, preventing it from escaping into space.

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- This natural process keeps Earth warm enough for life—but excess GHG emissions from fossil fuels cause **global warming**.

Evidence of Climate Change

- **Rising Sea Levels**: Due to melting glaciers and thermal expansion of oceans.
- **Melting Glaciers**: Himalayas retreating rapidly, threatening water supply to millions.
- **Heat Waves**: More frequent, longer, and intense (e.g., India 2022).
- **Erratic Monsoons**: Unpredictable rainfall patterns → floods in some regions, droughts in others.
- **IPCC Reports** confirm human-induced climate change as the main driver.

2. Impact of Climate Change

a) Natural Systems

1. **Coral Bleaching**: Warmer oceans expel algae (zooxanthellae) → corals turn white and die.
2. **Biodiversity Loss**: Habitats shrink, species unable to adapt → extinction risk rises.
3. **Desertification**: Land degradation due to heat and drought, reducing soil productivity.

b) Human Systems

1. **Agriculture Stress**
   - Changing rainfall patterns affect crop yield.
   - Example: Rice and wheat production in India faces major risks.
2. **Health Risks**
   - **Vector-borne diseases** (malaria, dengue) spread due to warmer, wetter climates.
   - Heat strokes and respiratory issues worsen in cities.
3. **Urban Flooding**
   - Intensified rainfall events overwhelm stormwater drains.
   - Example: Chennai Floods (2015).

c) Kerala/India Case Studies

1. **Kerala Floods (2018 & 2019)**

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- Linked to **erratic monsoon patterns** and high reservoir levels.
- Over 400 deaths in 2018, heavy damage to infrastructure and livelihoods.

2. **Heat Waves in North India**
   - Delhi, Rajasthan, Uttar Pradesh recorded **temperatures >47°C** in recent years.
   - Increased mortality among outdoor workers.
3. **Sea Level Rise Threatening Coastal Communities**
   - Kerala's coastal belts (Alappuzha, Kochi) and islands like Lakshadweep at risk.
   - Estimated 36 million Indians could face coastal flooding by 2050.

3. Engineering Solutions to Climate Change

Engineers have a **triple responsibility**: **mitigate emissions, enable adaptation, and build resilience**.

A. Mitigation – Reducing Greenhouse Gas Emissions

1. **Renewable Energy**
   - Large-scale solar, wind, hydro, biomass, and green hydrogen adoption.
   - Example: India's **International Solar Alliance (ISA)** initiative.
2. **Carbon Capture & Storage (CCS)**
   - Capturing CO_2 from industries and power plants → storing it underground.
3. **Energy Efficiency**
   - Smart grids, efficient appliances, green buildings with insulation and natural cooling.
   - Electric mobility and fuel-efficient engines.

B. Adaptation – Adjusting to New Climate Realities

1. **Climate-Resilient Crops**
   - Drought-resistant rice, flood-tolerant paddy varieties.
   - Bio-engineered crops to handle salinity in coastal areas.
2. **Flood-Proof Housing & Infrastructure**
   - Elevated housing, stilt construction in flood-prone regions.

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- Permeable pavements and rain gardens to reduce waterlogging.

3. **Rainwater Harvesting (RWH)**
    - Mandatory rooftop collection in cities to reduce water scarcity.
    - Restoring traditional water tanks and ponds.

C. Building Resilience – Long-Term Strategies

1. **Smart Cities with Sustainable Transport**
    - Integration of metro, BRT, e-mobility, and non-motorized transport.
    - Example: Kochi Metro + Water Metro integration.
2. **Green Infrastructure**
    - Urban forests, vertical gardens, green roofs to absorb CO_2 and reduce heat islands.
    - Mangrove restoration for coastal protection.
3. **Early Warning Systems**
    - Use of AI + IoT sensors for flood/tsunami alerts.
    - Improved meteorological forecasting to save lives and reduce damage.

4. Ethical and Professional Responsibility

- **Preventive Ethics**: Engineers must anticipate long-term climate impacts while designing.
- **Social Justice**: Ensure solutions benefit vulnerable groups (farmers, coastal dwellers, urban poor).
- **Intergenerational Responsibility**: Decisions today must not compromise the survival of future generations.

III. ENVIRONMENTAL POLICIES AND REGULATIONS

1. Key Environmental Policies

Environmental policies are legal and institutional frameworks created to safeguard ecosystems, reduce pollution, and ensure sustainable development. They operate at **national** and **international** levels.

A. Indian Environmental Policies

1. **Environmental Protection Act (1986)**

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- Enacted after the **Bhopal Gas Tragedy (1984)**.
- Serves as the "umbrella legislation" empowering the government to protect and improve environmental quality.
- Provides authority to:
  - Set emission and effluent standards.
  - Regulate industrial operations.
  - Protect ecologically sensitive areas.
- Example: Coastal Regulation Zone (CRZ) notifications to protect India's coastal ecosystems.

2. **National Action Plan on Climate Change (NAPCC) – 2008**
   - Framework for India's climate strategy.
   - Eight National Missions:
     - **National Solar Mission** (increase solar capacity).
     - **National Water Mission** (water efficiency).
     - **Energy Efficiency Mission**.
     - **Sustainable Habitat Mission**.
     - **Green India Mission** (afforestation).
   - State governments develop **State Action Plans on Climate Change (SAPCCs)**.
3. **Plastic Waste Management Rules (2016, amended 2021)**
   - Ban on certain **single-use plastics** (e.g., straws, cutlery, polythene bags <50 microns).
   - Promotes Extended Producer Responsibility (EPR): producers must collect and manage post-consumer plastic waste.
   - Encourages biodegradable and compostable alternatives.
4. **Renewable Energy Policies (MNRE initiatives)**
   - Ministry of New and Renewable Energy (MNRE) promotes solar, wind, biomass, small hydro, and green hydrogen.

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- Programs: **Soura rooftop solar scheme (Kerala)**, **National Wind-Solar Hybrid Policy (2018)**.
- Target: **500 GW of renewable capacity by 2030**.

B. International Environmental Agreements

1. **Paris Agreement (2015)**
   - A landmark UN climate treaty signed by 196 countries.
   - Aim: Limit global warming to **well below 2°C**, preferably 1.5°C, compared to pre-industrial levels.
   - Countries submit **Nationally Determined Contributions (NDCs)** – India pledged to reduce emission intensity by 45% by 2030.
2. **Montreal Protocol (1987)**
   - International treaty to phase out **ozone-depleting substances (ODS)** such as chlorofluorocarbons (CFCs).
   - Widely regarded as the most successful environmental treaty.
   - Helped in recovery of the ozone layer, preventing millions of skin cancer cases.
3. **UN Sustainable Development Goals (SDGs – 2015)**
   - 17 global goals to be achieved by 2030.
   - **Goal 7**: Affordable & Clean Energy.
   - **Goal 13**: Climate Action.
   - **Goal 15**: Life on Land.
   - SDGs provide a holistic framework integrating **environmental, social, and economic sustainability**.

2. Role of Engineers in Policy and Compliance

Engineers bridge the gap between **policy framework** and **practical implementation**.

1. **Designing Technologies that Comply with Environmental Norms**
   - Developing cleaner production processes.
   - Using eco-friendly materials and minimizing emissions.

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- Example: Electric vehicle engineers designing systems compliant with **Bharat Stage VI (BS-VI) emission norms**.

2. **Conducting Environmental Impact Assessments (EIA)**
    - A mandatory process for major projects (dams, industries, highways).
    - Identifies potential environmental and social impacts.
    - Proposes mitigation measures before project approval.
3. **Ensuring Compliance with Environmental Standards (ISO 14001)**
    - ISO 14001 provides a framework for **Environmental Management Systems (EMS)**.
    - Engineers ensure continuous monitoring, waste reduction, and sustainable resource use.
4. **Ethical Responsibility of Engineers**
    - Engineers must **resist corporate or political pressure** to bypass safety and environmental laws.
    - Example: **Volkswagen Emissions Scandal (2015)** – unethical engineering choices led to massive global penalties.

3. Ethical Considerations in Environmental Policy

1. Justice

- Vulnerable communities (fishermen, indigenous people, slum dwellers) often suffer the most from pollution, floods, and climate change.
- Engineers have a duty to design solutions that **protect marginalized groups**.
- Example: Resettling communities displaced by dams with dignity and livelihood security.

2. Preventive Ethics

- Anticipating harm before it occurs rather than reacting afterward.
- Example: Proper **site selection** for chemical industries away from residential areas to avoid another Bhopal disaster.
- Designing renewable projects that minimize ecological harm (e.g., avoiding solar parks in biodiversity hotspots).

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3. Intergenerational Responsibility

- Today's actions must not compromise the ability of **future generations** to meet their needs (as per the Brundtland Report, 1987).
- Example: Over-extraction of groundwater today risks water scarcity for future generations.
- Engineers must ensure **sustainable use of resources** such as forests, rivers, and minerals.

IV. CASE STUDIES AND FUTURE DIRECTIONS

1. Case Studies

1.1 Solar Parks in Rajasthan – India's Renewable Energy Hub

- Rajasthan has abundant solar radiation and vast arid land, making it ideal for large-scale solar deployment.
- **Bhadla Solar Park** (Jodhpur, Rajasthan)
  - One of the **largest solar parks in the world**, capacity > 2,245 MW.
  - Supplies clean energy to multiple states, reducing dependence on coal.
- Significance:
  - Helps India move towards its **450 GW renewable energy target by 2030**.
  - Demonstrates economies of scale in solar deployment.
- Challenges:
  - Land conflicts with local communities.
  - Dust storms reduce panel efficiency.

1.2 Kochi Water Metro – World's First Solar-Powered Water Metro

- Kerala launched the **Kochi Water Metro** in 2023.
- Uses **solar-powered electric hybrid ferries**, connecting 10 islands around Kochi city.
- Features:
  - Reduces road congestion.
  - Uses clean solar energy for operation.

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- Promotes **integrated multimodal transport** with Kochi Metro Rail.

- Ethical & Sustainable Dimensions:
  - First of its kind globally, showcasing India's innovation.
  - Provides **affordable and eco-friendly transport** for urban island communities.

1.3 Germany's Energiewende – Energy Transition

- **Energiewende** ("energy transition") is Germany's long-term policy to shift from fossil fuels and nuclear power → renewable energy.
- Goals:
  - Reduce GHG emissions by **65% by 2030** (compared to 1990).
  - Phase out nuclear by 2023, coal by 2038.
  - Achieve **80% renewables in electricity mix by 2030**.
- Successes:
  - Huge growth in wind and solar capacity.
  - Creation of green jobs and technology leadership.
- Challenges:
  - High cost of electricity.
  - Grid integration and storage issues.

1.4 Bhutan – A Carbon-Negative Nation

- Bhutan absorbs more CO_2 than it emits.
- **Why?**
  - Over **70% forest cover** is constitutionally protected.
  - Hydropower exports (mainly to India) provide clean energy revenues.
  - Strong government policies aligning with **Gross National Happiness (GNH)**.
- Lesson for Engineers:
  - Sustainable development is possible by balancing energy, economy, and environment.
  - Conservation and renewable energy can coexist.

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2. Emerging Trends

2.1 Green Hydrogen Economy

- Hydrogen produced via **electrolysis using renewable energy**.
- Uses: Fuel for industries (steel, cement, refineries), transport (buses, trucks, ships).
- India's **National Green Hydrogen Mission (2023)** targets 5 MMT annual production by 2030.
- Example: Japan investing in hydrogen-powered cars and refueling infrastructure.

2.2 Electric Aviation & Hydrogen-Powered Ships

- **Electric Aviation**: Small electric planes under development for regional travel (e.g., Norway's plan for all short-haul flights to be electric by 2040).
- **Hydrogen Ships**: Used in Europe for cargo and passenger ferries.
- Potential: Decarbonizing hard-to-electrify transport sectors (aviation, shipping).

2.3 Circular Economy in Renewable Energy

- Focus on recycling and reuse of renewable infrastructure:
  - **Battery Recycling**: Recovering lithium, cobalt, nickel from used EV batteries.
  - **Wind Blade Reuse**: Composite blades repurposed for construction materials.
  - **Solar Panel Recycling**: Extracting silicon and rare metals.
- Ensures renewable energy technologies themselves do not create future waste problems.

2.4 Carbon Capture & Storage (CCS)

- Captures CO_2 from industrial plants (steel, cement, thermal power).
- Stores it in geological formations or reuses it in synthetic fuels.
- Example: Sleipner CCS Project (Norway) has stored 1 million tonnes of CO_2 annually since 1996.
- India's challenge: High costs, but CCS may be essential for **net-zero 2070 goal**.

3. Future Directions

3.1 Engineers as Moral Leaders in Sustainability

- Engineers must go beyond technical fixes → act as **ethical leaders**.

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- Balance economic growth with environmental justice.
- Resist pressures from industry/politics to dilute environmental standards.

3.2 Integrating AI and IoT in Smart Grids

- Smart grids use AI to forecast demand, manage renewable fluctuations, and reduce wastage.
- IoT devices (smart meters, sensors) enable real-time monitoring.
- Improves energy efficiency and reduces blackouts.
- Example: India's Smart Meter National Program (250 million smart meters by 2025).

3.3 Climate-Resilient Infrastructure

- Engineers must design cities and infrastructure to withstand climate shocks.
- Examples:
    - **Elevated metro lines** in flood-prone regions.
    - **Cyclone-resistant housing** in coastal Odisha.
    - **Urban wetlands restoration** to reduce flooding.

3.4 International Collaborations for Technology Transfer

- Rich nations have advanced clean tech; developing nations have implementation potential.
- Collaboration ensures **equity in global climate action**.
- Example: India–France partnership in International Solar Alliance (ISA).

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Question Bank

1. List any three renewable energy sources.
2. Differentiate between photovoltaic (PV) and solar thermal systems.
3. What is the difference between onshore and offshore wind turbines?
4. Mention two advantages of biomass as an energy source.
5. Define "smart grid" in renewable energy systems.
6. What are hybrid renewable energy systems? Give one example.
7. State two challenges in large-scale deployment of solar energy.
8. Define climate. How does it differ from weather?
9. Name any three greenhouse gases responsible for global warming.
10. Mention two pieces of evidence for climate change.
11. Give two examples of climate change impacts on agriculture.
12. Write a short note on "urban flooding" with one Indian example.
13. List any two health risks caused by climate change.
14. State two examples of climate change impacts in Kerala.
15. State the main objective of the Environmental Protection Act (1986).
16. Mention any two missions under India's National Action Plan on Climate Change (NAPCC).
17. What is Extended Producer Responsibility (EPR) in waste management?
18. Write a short note on the Paris Agreement (2015).
19. What is ISO 14001?
20. Name two international environmental agreements.
21. State one key feature of the Plastic Waste Management Rules (2016).
22. Name the largest solar park in India and its location.
23. Which is the world's first solar-powered water metro project?
24. Define Energiewende.
25. How is Bhutan a carbon-negative country?
26. What is green hydrogen?
27. Mention one challenge of recycling wind turbine blades.
28. Give two examples of future sustainable technologies.
29. Explain the working principles of solar PV systems and their applications.
30. Compare the advantages and limitations of small-scale vs. large-scale hydropower projects.
31. Discuss the role of energy storage technologies (battery banks, pumped hydro) in renewable integration.
32. Evaluate the opportunities and challenges for India in achieving 450 GW renewable energy target by 2030.
33. Case Study Question: Explain how Kochi Water Metro demonstrates the integration of renewable energy into sustainable transport.
34. Explain the greenhouse effect with a neat diagram.
35. Discuss the impacts of climate change on natural and human systems with examples.
36. Explain the causes and consequences of Kerala floods (2018 & 2019).
37. Evaluate the engineering solutions for mitigation, adaptation, and resilience to climate change.
38. Discuss the ethical responsibilities of engineers in addressing climate change.
39. Explain the key features of India's Environmental Protection Act (1986).
40. Discuss the importance and challenges of the National Action Plan on Climate Change (NAPCC).

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41. Write detailed notes on:
    a. Paris Agreement (2015)
    b. Montreal Protocol

42. Explain the role of engineers in environmental policy compliance with examples.
43. Critically analyze the ethical considerations in environmental policy with reference to justice, preventive ethics, and intergenerational responsibility.
44. Explain any two case studies in renewable energy adoption (e.g., Rajasthan solar parks, Kochi Water Metro).
45. Discuss Germany's Energiewende policy and its relevance to India.
46. Explain the concept of a green hydrogen economy and its potential in India.
47. Analyze how circular economy principles can be applied to renewable energy systems (battery recycling, wind blade reuse).
48. Future Essay Question: Evaluate the role of engineers as moral leaders in sustainability and how AI/IoT can shape the energy future.

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